

Bag of Words (BoW)

The Complete A–Z Tutorial

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1. What Is Bag of Words?

Bag of Words (BoW) is the simplest way to turn text into numbers so a machine learning model can use it. The core idea: **ignore grammar and word order, and represent a document purely by which words appear and how often.**

Think of it like literally dumping all the words of a sentence into a bag — you can count how many of each word are in there, but you can't tell what order they were in.

"the movie was not good, it was bad" vs "the movie was not bad, it was good"

Under BoW, these two sentences (which mean opposite things) produce **identical vectors**, same words, same counts. This is BoW's central limitation, and it's important to understand it from the start because everything else in this tutorial builds around working with or around that weakness.

2. The Mathematical Formulation

Given a corpus (collection of documents) $D = \{d_1, d_2, \dots, d_n\}$

1. Build a **vocabulary** $V = \{w_1, w_2, \dots, w_n\}$, the set of all unique tokens across the corpus.
2. Represent each document d_i as a vector $v_i \in \mathbb{R}^m$, where each dimension corresponds to one vocabulary word.
3. The value at position j is typically the **count** of word w_j in document d_i (or just 1/0 for binary BoW).

The result is a document term matrix: rows = documents, columns = vocabulary word, values = counts.

Worked example

Corpus:

- d1: "the cat sat on the mat"
- d2: "the dog sat on the log"

Vocabulary (alphabetical): [cat, dog, log, mat, on, sat, the]

doc	cat	dog	log	mat	on	sat	the
d1	1	1	0	1	1	1	2
d2	0	1	1	0	1	1	2

Each row is now a numeric vector a classifier can consume.

3. The Preprocessing Pipeline (before vectorization)

Raw text is messy. The standard pipeline, in order:

1. **Lowercasing** — "Movie" and "movie" should map to the same token.
2. **Noise removal** — strip HTML tags, URLs, punctuation, special characters, numbers (task-dependent).
3. **Tokenization** — split text into individual words/tokens.
4. **Stopword removal** — drop high-frequency, low-information words (**the, is, a, and...**).
Optional — for sentiment tasks, be careful: words like "not" and "no" are often stopwords by default but carry sentiment-critical negation. Removing them can hurt accuracy.
5. **Stemming or Lemmatization** — collapse word variants to a root form:
 - **Stemming** (e.g., Porter Stemmer): crude, rule-based chopping. **"running"** → **"run"**, but also **"studies"** → **"studi"** (not a real word). Fast, slightly noisy.
 - **Lemmatization**: dictionary/grammar-aware, returns real words (**"better"** → **"good"**). More accurate, slower, needs POS context for best results.
6. (Optional) **Spelling correction / contraction expansion** — **"don't"** → **"do not"**.

Order matters: tokenize before removing stopwords/stemming; lowercase before matching against a stopwords list.

4. Building the Vocabulary

The vocabulary is the set of unique tokens that become your feature columns. Key controls (as in `sklearn.CountVectorizer`):

- **max_features** — keep only the top-N most frequent words. Controls dimensionality directly.
- **min_df** — ignore terms that appear in fewer than N documents (or < a proportion). Removes rare/noisy words, typos, and near-unique tokens that don't generalize.
- **max_df** — ignore terms that appear in more than N documents (or > a proportion). Removes overly common words that behave like stopwords even if not in a stopwords list.
- **ngram_range** — see below (Section 5).

Critical rule to avoid data leakage: fit the vocabulary **only on the training set**, then use `.transform()` (not `.fit_transform()`) on validation/test data. If you fit on the full dataset before splitting, information about test-set word distributions leaks into training — this is one of the most common beginner mistakes in the BoW pipeline.

5. Variants of Bag of Words

Variant	Description	When to use
Binary BoW	1 if word present, 0 if not (ignores count)	Short texts where repetition isn't meaningful (e.g., tweets)
Count BoW	Raw frequency counts	Default; works when word frequency carries signal
N-gram BoW	Includes multi-word sequences (bigrams, trigrams) as features, not just single words	Captures some local word order/context, e.g. " not good " becomes its own feature instead of being lost
TF-IDF (technically a BoW variant, not vanilla BoW)	Weights counts by how distinctive a word is across the corpus (downweights common words, upweights rare-but-informative ones)	Almost always outperforms raw-count BoW for classification

N-grams example: with `ngram_range=(1,2)` on "not good", you get features `not`, `good`, and `not_good` — this partially recovers the negation information that pure unigram BoW throws away.

6. Sparsity and Storage

For a vocabulary of 20,000 words, a document with 50 unique words has a vector that's 99.75% zeros. This is why BoW matrices are stored as **sparse matrices** (e.g., `scipy.sparse.csr_matrix`), not dense NumPy arrays — storing all those zeros explicitly would waste enormous memory.

- A 25,000-document corpus × 20,000-word vocabulary as a dense `float64` array ≈ 4 GB.
- As a sparse matrix storing only non-zero entries ≈ a few MB.

Practical implication: some models handle sparse input natively and efficiently (Logistic Regression, LinearSVC, KNN, Naive Bayes, Random Forest, Decision Tree, XGBoost). Others (Naive Bayes (GaussianNB), dense ANN input layers) may require `.toarray()`, which can blow up memory — this is a real bottleneck to plan for, not just a technical footnote.

7. Implementation (Python / scikit-learn)

```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import train_test_split

texts = ["I loved this movie", "This movie was terrible", "Great acting and plot"]

labels = [1, 0, 1]

X_train_text, X_test_text, y_train, y_test = train_test_split(texts,
                                                                labels,
                                                                test_size=0.2,
                                                                random_state=42)

vectorizer = CountVectorizer(lowercase=True, stop_words='english',
                              ngram_range=(1, 2), max_features=10000, min_df=2, max_df=0.9, binary=False
                              # True = binary BoW)

# Fit ONLY on training data
X_train = vectorizer.fit_transform(X_train_text)

# Transform test data using the SAME fitted vocabulary
```

```
X_test = vectorizer.transform(X_test_text)

print(X_train.shape) # (n_docs, vocab_size)

print(vectorizer.get_feature_names_out()[:20]) # inspect vocabulary
```

Saving/loading the fitted vectorizer (needed for inference later):

```
import pickle

with open('bow_vectorizer.pkl', 'wb') as f:
    pickle.dump(vectorizer, f)

# later / in production

with open('bow_vectorizer.pkl', 'rb') as f:
    vectorizer = pickle.load(f)

X_new = vectorizer.transform(["a brand new review"])
```

8. Which Models Pair Well With BoW

- **Naive Bayes** — theoretically closest fit; assumes conditional independence between words, which is exactly the assumption BoW already makes. Fast, strong baseline.
 - **Logistic Regression / LinearSVC** — excellent with high-dimensional sparse data; typically among the top performers on BoW/TF-IDF text classification.
 - **Tree-based models (Decision Tree, Random Forest, XGBoost)** — can work, but tend to plateau earlier as vocabulary size grows, since trees split on one feature at a time and struggle to exploit thousands of sparse, mostly-zero features efficiently.
 - **KNN** — usually weak on BoW; distance metrics degrade badly in high-dimensional sparse spaces (curse of dimensionality).
 - **ANN (feed-forward)** — can work well and keep improving as vocabulary grows, but needs dense input, which raises memory considerations at scale, and needs regularization (dropout, early stopping) to avoid overfitting the huge input dimensionality.
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9. Advantages of BoW

- Simple to understand, implement, and debug.
 - Cheap computationally compared to embeddings or transformers.
 - Works surprisingly well as a strong baseline for topic classification, spam detection, and sentiment analysis.
 - Fully interpretable — you can inspect exactly which words drove a prediction (especially with linear models, via coefficients).
 - No need for pretrained resources or external data — built entirely from your own corpus.
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10. Limitations of BoW

- **No word order** — "not good" and "good, not..." can look identical or very different from their real meaning.
 - **No semantics** — "excellent" and "great" are treated as completely unrelated dimensions, even though they're near-synonyms. BoW has no notion of meaning or similarity between words.
 - **High dimensionality & sparsity** — vocabularies can reach tens of thousands of features, most of them irrelevant to any single document.
 - **Out-of-vocabulary (OOV) problem** — any word not seen during training is simply dropped at inference time; the model has no way to represent it.
 - **No context sensitivity** — the word "bank" gets one vector-dimension whether it means a riverbank or a financial institution.
 - **Frequency bias** — very common words dominate raw counts unless you control for it (this is exactly the problem TF-IDF was designed to fix).
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11. BoW vs TF-IDF vs Word2Vec vs Transformers

Method	Captures word order?	Captures meaning / semantics?	Dimensionality	Typical use case
BoW (counts)	No	No	High, sparse	Fast baseline, interpretable models
TF-IDF	No	No (but reweights by importance)	High, sparse	Usually beats raw BoW for classification
Word2Vec / GloVe	No (per-word), some via window context during training	Yes — similar words have similar vectors	Low, dense (e.g. 100–300 dims)	When semantic similarity matters
RNN/LSTM on embeddings	Yes (sequential)	Yes	Low, dense	When order and context both matter
Transformers (BERT etc.)	Yes (full context)	Yes, contextual (same word gets different vectors per context)	Low, dense	State of the art, most compute intensive

The general trend as you move down this table: representations get denser, smaller, and more semantically meaningful — at the cost of more compute and less direct interpretability.

12. Common Pitfalls (and how to avoid them)

1. **Fitting the vectorizer on the full dataset before train/test split** → data leakage.
Always `fit_transform` on train, `transform` on test/val.
 2. **Removing negation words as "stopwords"** for sentiment tasks → destroys the exact signal you need. Inspect your stopwords list before applying it blindly.
 3. **Not controlling vocabulary size** → memory blowup, overfitting, slow training. Use `max_features/min_df/max_df` deliberately, and treat them as hyperparameters to tune rather than fixed defaults.
 4. **Converting sparse matrices to dense (`.toarray()`) unnecessarily** → can exhaust RAM for little benefit; only do it when a model strictly requires dense input.
 5. **Comparing models unfairly** — if one model was tuned with a different vocabulary size or preprocessing than another, the comparison is invalid. Keep preprocessing identical across models unless you're specifically testing preprocessing choices.
 6. **Assuming more vocabulary is always better** — tree-based models often plateau early (e.g., around 10K features), while linear/ANN models may keep improving through 20-30K+; check this empirically rather than assuming.
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13. When to Use BoW (and When to Move On)

BoW is a good choice when:

- You need a fast, interpretable baseline.
- The task is largely about *topic* or *keyword presence* (e.g., spam detection, topic tagging).
- You have limited compute/data and can't justify training or fine-tuning embeddings/transformers.

Move beyond BoW when:

- Word order and negation meaningfully change the label (nuanced sentiment, sarcasm).
- You need the model to generalize to synonyms/paraphrases it never saw in training.
- You have enough data and compute to benefit from dense embeddings (Word2Vec/GloVe) or contextual models (BERT-family).

In practice, BoW/TF-IDF are the right starting point for almost any text classification project — they establish a baseline number quickly, and you only move to embeddings/transformers once you've confirmed the extra complexity actually buys you accuracy.

14. Quick Reference Checklist

- ☐ Clean text (lowercase, strip noise)
- ☐ Tokenize
- ☐ Decide: remove stopwords? (careful with negation for sentiment)
- ☐ Decide: stem or lemmatize?
- ☐ Split into train/test **before** fitting vectorizer
- ☐ Choose: binary vs count vs n-grams vs TF-IDF
- ☐ Set **max_features** / **min_df** / **max_df** deliberately
- ☐ Fit vectorizer on train only, transform test with same vectorizer
- ☐ Save the fitted vectorizer (pickle) alongside the trained model
- ☐ Pick a model suited to sparse, high-dimensional input
- ☐ Compare vocabulary-size sensitivity across model types